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Simulation and Analysis of BB84 Quantum Key Distribution for Secure Data Transmission

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Acknowledgement

I hereby certify that the work which is being presented in the project report entitled **"Simulation and Analysis of BB84 Quantum Key Distribution for Secure Data Transmission"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr Sikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. Abstract

Secure communication is a critical requirement in modern digital networks, particularly with the increasing computational capabilities that threaten classical cryptographic techniques. Quantum Key Distribution (QKD) provides an alternative approach to secure key exchange by exploiting the principles of quantum mechanics. Among various QKD schemes, the BB84 protocol remains one of the most widely studied and implemented methods for detecting eavesdropping during transmission.

This work presents a simulation-based performance and security analysis of the BB84 protocol under varying channel noise conditions and intercept-resend attack scenarios. A Python-based model was developed to simulate the interaction between the communicating parties and a potential adversary. The protocol behavior was evaluated using two performance metrics: Quantum Bit Error Rate (QBER) and key generation rate. Experimental results demonstrate that, in the absence of an attacker, QBER increases approximately linearly with channel noise, while the presence of an eavesdropper introduces a baseline error rate of nearly 25%, enabling reliable detection of interception. Additionally, the key generation rate decreases with increasing noise and adversarial interference, indicating reduced communication efficiency under disturbed channel conditions.

The results validate theoretical expectations of the BB84 protocol and highlight the effectiveness of simulation-based analysis in studying quantum cryptographic security mechanisms.

2. Introduction

With the rapid growth of digital communication systems and internet-based services, secure data transmission has become an essential requirement in modern computing environments. Conventional cryptographic techniques such as RSA and Elliptic Curve Cryptography rely on mathematical complexity to protect information. However, the emergence of quantum computing threatens these classical encryption techniques because quantum algorithms, such as Shor's algorithm, can solve factorization and discrete logarithm problems efficiently.

To address this limitation, quantum cryptography introduces a fundamentally new approach to secure communication based on the principles of quantum mechanics. Unlike classical cryptographic methods, quantum cryptography does not depend on computational hardness assumptions but instead relies on physical laws such as the uncertainty principle and the no-cloning theorem to ensure security.

Quantum Key Distribution (QKD) is one of the most important applications of quantum cryptography. It enables two communicating parties to securely generate a shared secret key while detecting the presence of an eavesdropper. Among various QKD protocols proposed in the literature, the BB84 is the first and most widely studied protocol.

The BB84 protocol allows a sender (Alice) and a receiver (Bob) to exchange cryptographic keys using quantum states encoded in different polarization bases. If an attacker (Eve) attempts to intercept the transmitted quantum states, the disturbance introduced during measurement can be detected through an increase in error rates.

Although the theoretical foundation of BB84 has been extensively studied, simulation-based analysis provides an effective way to understand its behaviour under realistic communication conditions such as channel noise and adversarial interference. In this work, a Python-based simulation framework is developed to evaluate the performance of the BB84 protocol under varying noise levels and intercept-resend attacks.

The performance of the protocol is analysed using two key metrics:

- Quantum Bit Error Rate (QBER)
- Key Generation Rate

The experimental results demonstrate how the BB84 protocol maintains security by detecting disturbances introduced during transmission.

3. Methodology

The methodology adopted in this project involves developing a simulation-based implementation of the BB84 protocol using Python and evaluating its performance under different communication scenarios.

The system model includes three entities:

- Alice (sender)
- Bob (receiver)
- Eve (eavesdropper)

The simulation begins with Alice generating a random sequence of binary bits that represent the initial key. For each bit, Alice randomly selects an encoding basis from two possible polarization bases:

- Rectilinear basis (+)
- Diagonal basis (×)

These bases correspond to different quantum state representations of photons.

After encoding, the quantum states are transmitted through a simulated quantum communication channel to Bob. Bob independently selects measurement bases randomly for each received qubit. If Bob's chosen basis matches Alice's encoding basis, the measured bit matches the transmitted bit with high probability. Otherwise, the measurement result becomes random due to the principles of quantum mechanics.

Once transmission is completed, Alice and Bob compare their selected bases over a classical communication channel. They discard all bits where their bases do not match. The remaining bits form the sifted key.

To simulate adversarial conditions, an intercept-resend attack model is implemented. In this attack, Eve intercepts each transmitted qubit and measures it using a randomly chosen basis. Since Eve does not know Alice's encoding basis, incorrect measurement introduces disturbances

in the quantum states. After measurement, Eve retransmits the qubit to Bob. This process increases the probability of errors in the final shared key.

In addition to adversarial interference, channel noise is also introduced into the simulation. Noise is modelled as a probabilistic bit-flip process in which each transmitted bit has a predefined probability of being inverted. This simulates real-world imperfections in communication channels.

To evaluate system performance, two important metrics are calculated:

- Quantum Bit Error Rate (QBER)
- Key Generation Rate

Multiple simulation trials are conducted across different noise levels to ensure statistical reliability of the results.

4. Tools And Technologies

The implementation of the simulation model was carried out using modern programming tools and libraries suitable for scientific experimentation.

4.1. Programming Language

Python was selected as the primary programming language due to its simplicity, flexibility, and strong support for simulation-based modelling.

4.2. Libraries Used

The following Python libraries were used during implementation:

- random library for generating random bits and bases
- matplotlib library for plotting graphs and visualizing results

4.3. Development Environment

The simulation was implemented and executed using Jupyter Notebook, which provides an interactive environment for testing and visualization.

4.4. Conceptual Framework

The project is based on the following theoretical concepts:

- Quantum Key Distribution (QKD)
- BB84 protocol
- Intercept-resend attack model
- Channel noise modelling
- Error rate estimation techniques

These tools and technologies enabled efficient modelling of the BB84 protocol and visualization of experimental outcomes.

5. Implementation

The implementation of the BB84 quantum key distribution protocol in this project was carried out using a simulation-based approach in Python. Since physical quantum communication hardware is not easily accessible in a standard academic environment, a classical simulation model was developed to replicate the behaviour of quantum transmission, measurement processes, channel noise, and adversarial interception. The objective of the implementation was to reproduce the logical workflow of the BB84 protocol and experimentally evaluate its performance under controlled conditions.

The simulation models three primary participants involved in the communication process:

- Alice (the sender)
- Bob (the receiver)
- Eve (the eavesdropper)

Each component of the protocol was implemented step by step to closely resemble the theoretical structure of BB84.

5.1. Random Bit Generation by Alice

The first stage of implementation involves the generation of a random binary sequence by Alice. This sequence represents the original secret key that Alice intends to share securely with Bob. The randomness of this sequence is essential for ensuring cryptographic security.

In the simulation, Python's built-in random number generator was used to generate binary values (0 and 1) with equal probability. A fixed number of bits was generated for each experiment to maintain consistency across simulation trials. In this study, the number of transmitted bits was set to 1000 for each iteration of the experiment.

Mathematically, the bit sequence generated by Alice can be represented as:

$B_A = \{b_1, b_2, b_3, \dots, b_n\}$

where each bit $b_i \in \{0,1\}$

The randomness of this sequence ensures that the generated key cannot be predicted by an external attacker.

5.2. Selection of Encoding Bases

After generating the binary sequence, Alice selects encoding bases randomly for each bit. In the BB84 protocol, two polarization bases are used:

- Rectilinear basis (+)
- Diagonal basis (×)

Each basis represents a different method of encoding quantum states. The rectilinear basis corresponds to horizontal and vertical polarizations, while the diagonal basis corresponds to diagonal polarization states.

For each bit generated, Alice randomly selects one of these two bases using equal probability. The sequence of bases selected by Alice is represented as:

$$S_A = \{s_1, s_2, s_3, \dots, s_n\}$$

where each $s_i \in \{+, \times\}$

This step simulates the preparation of quantum states before transmission through the communication channel.

5.3. Transmission Through the Quantum Channel

Once the bits and bases are selected, Alice encodes the binary values into quantum states according to the chosen bases. These encoded qubits are then transmitted to Bob through a simulated quantum communication channel

Since this implementation is simulation-based rather than hardware-based, quantum state transmission is modeled logically instead of physically. The encoding process determines how Bob's measurement outcome will depend on the compatibility between Alice's encoding basis and Bob's measurement basis.

The simulation assumes that if Bob selects the same basis as Alice, the transmitted bit is received correctly. If the bases differ, the measurement outcome becomes random due to the uncertainty principle of quantum mechanics.

5.4. Random Measurement Basis Selection by Bob

Upon receiving the transmitted qubits, Bob independently selects measurement bases for each qubit. Similar to Alice's encoding process, Bob randomly chooses between the rectilinear and diagonal bases.

The measurement basis sequence selected by Bob can be represented as:

$$S_B = \{s'_1, s'_2, s'_3, \ldots, s'_n\}$$

For each received qubit:

- If $s_i = s'_i$, Bob obtains the correct bit
- If $s_i \neq s'_i$, Bob obtains a random bit

This behavior simulates the probabilistic nature of quantum measurement.

5.5. Simulation of Intercept–Resend Attack

To evaluate the security of the BB84 protocol under adversarial conditions, an intercept–resend attack model was implemented. In this scenario, an attacker named Eve intercepts each transmitted qubit before it reaches Bob.

Eve performs the following operations:

- Randomly selects a measurement basis
- Measures the intercepted qubit
- Resends the measured qubit to Bob

Since Eve does not know Alice's encoding basis, she chooses the correct basis only 50% of the time. When the wrong basis is selected, the quantum state collapses into an incorrect value, introducing detectable disturbances in the transmission.

This attack model increases the probability of errors in the shared key and allows Alice and Bob to detect Eve's presence by monitoring the Quantum Bit Error Rate.

The intercept–resend process was implemented programmatically by introducing an intermediate measurement stage between Alice and Bob whenever the adversarial condition was enabled.

5.6. Modeling Channel Noise

In practical communication systems, noise is an unavoidable factor that affects signal transmission reliability. To simulate realistic communication conditions, channel noise was incorporated into the implementation using a probabilistic bit-flip model.

In this model, each transmitted bit has a predefined probability of being inverted during transmission. The noise probability parameter was varied between 0 and 0.20 in incremental steps to observe its effect on system performance.

Mathematically, noise can be represented as:

$$P(\text{bit flip}) = p$$

where p represents the noise probability.

If a randomly generated value falls below the specified probability threshold, the transmitted bit is flipped. This mechanism simulates environmental disturbances and imperfections in real quantum communication channels.

5.7. Key Sifting Process

After transmission is completed, Alice and Bob compare their selected bases over a classical communication channel. This step is known as the sifting process.

During sifting:

- Bits corresponding to mismatched bases are discarded
- Bits corresponding to matched bases are retained

The remaining bits form the sifted key.

Mathematically:

$$K = \{b_i \mid s_i = s_i'\}$$

This process ensures that only reliable measurements contribute to the final shared key.

5.8. Quantum Bit Error Rate Calculation

To evaluate the reliability of the communication channel and detect potential eavesdropping, the Quantum Bit Error Rate (QBER) was calculated.

QBER is defined as the ratio of incorrect bits to the total number of sifted bits:

$$QBER = \frac{\text{Number of mismatched bits}}{\text{Total sifted bits}}$$

A higher QBER indicates greater disturbance in the communication channel, which may result from noise or adversarial interception.

In the simulation, QBER was computed after the sifting process by comparing Alice's and Bob's retained key sequences.

5.9. Key Generation Rate Calculation

In addition to QBER, the key generation rate was used as a performance metric to evaluate communication efficiency.

The key generation rate represents the fraction of correctly shared bits after transmission and sifting relative to the total number of transmitted bits.

It is calculated as:

$$Key\ Rate = \frac{\text{Number of correct shared bits}}{\text{Total transmitted bits}}$$

This metric provides insight into how efficiently the protocol generates usable secret keys under different channel conditions.

5.10. Experimental Iterations and Averaging

To ensure statistical reliability of the simulation results, each experiment was repeated multiple times for every noise level. In this implementation, each configuration was executed 30 times and the average values of QBER and key generation rate were recorded.

This averaging process reduces fluctuations caused by randomness and produces more stable and accurate experimental observations.

Noise probability values ranging from 0 to 0.20 were tested systematically to analyze the relationship between channel noise and protocol performance under both normal and adversarial conditions.

5.11. Graphical Visualization of Results

The final stage of implementation involved plotting graphs to visualize experimental outcomes. The matplotlib library was used to generate the following plots:

- Noise vs QBER (without attacker)
- Noise vs QBER (with attacker)
- Noise vs key generation rate (without attacker)
- Noise vs key generation rate (with attacker)

These graphs clearly illustrate how channel noise and adversarial interference influence the reliability and efficiency of the BB84 protocol.

Visualization of results plays an important role in understanding protocol behavior and validating theoretical expectations through experimental simulation.

6. Major Findings

The simulation results provide several important insights into the behavior of the BB84 protocol under noisy and adversarial conditions.

First, in the absence of an eavesdropper, the Quantum Bit Error Rate increases approximately linearly with channel noise. This indicates that the protocol behaves predictably under normal communication conditions and errors are primarily due to channel imperfections.

Second, when an intercept-resend attacker is introduced, the QBER increases significantly even in the absence of channel noise. The baseline QBER observed in this case is approximately 25 percent, which matches theoretical expectations derived from quantum measurement principles.

Third, the key generation rate decreases steadily as channel noise increases. This occurs because higher noise levels introduce more transmission errors, reducing the number of usable bits in the final key.

Fourth, the presence of an eavesdropper further reduces the key generation rate compared to normal transmission conditions. This confirms that adversarial interference affects both security and efficiency of the communication system.

These observations demonstrate that the BB84 protocol effectively detects disturbances introduced by unauthorized interception.

7. Results

The performance of the simulated BB84 quantum key distribution system was evaluated under varying channel noise conditions and adversarial interference using an intercept–resend attack model. The experiments were conducted by transmitting 1000 bits per simulation run across multiple trials and averaging the obtained values to reduce statistical variation. The evaluation focused primarily on two performance metrics: the Quantum Bit Error Rate (QBER) and the key generation rate.

The first set of experiments analyzed the behavior of QBER under normal communication conditions without the presence of an eavesdropper. The results demonstrate that the QBER increases approximately linearly with channel noise. This behavior is consistent with theoretical expectations because, in the absence of adversarial interference, the only source of transmission error is the probabilistic noise introduced into the communication channel. As noise probability increases, the likelihood of bit flips also increases proportionally, resulting in a predictable rise in the error rate.

In the second scenario, the simulation included an intercept–resend attacker. The results showed a significant increase in QBER even when channel noise was absent. Specifically, the baseline QBER observed in the presence of the attacker was approximately 25 percent. This outcome aligns closely with theoretical predictions of the BB84 protocol, where an eavesdropper selecting incorrect measurement bases introduces detectable disturbances into the transmitted quantum states. As channel noise increased further, the QBER increased beyond the baseline level, indicating the combined impact of both environmental disturbances and adversarial activity.

A comparison between the normal communication scenario and the adversarial scenario clearly illustrates that the presence of an attacker produces a distinguishable increase in error rates. This difference confirms one of the fundamental security properties of the BB84 protocol, namely that eavesdropping attempts can be detected through monitoring of transmission errors.

In addition to QBER analysis, the key generation rate was evaluated to assess the efficiency of the protocol under different communication conditions. The key generation rate represents the proportion of usable shared bits after transmission and sifting relative to the total number of transmitted bits. Experimental observations showed that the key generation rate decreases

gradually as channel noise increases. This reduction occurs because higher noise levels increase the number of corrupted bits, thereby reducing the number of usable bits retained after the sifting process.

When an intercept–resend attacker was introduced, the key generation rate decreased further even in the absence of channel noise. This demonstrates that adversarial interference affects not only the security of the protocol but also its efficiency. As noise and eavesdropping effects accumulate, the number of correctly shared bits decreases significantly, reducing the effectiveness of key generation.

Graphical visualization of the results confirmed that the BB84 protocol maintains predictable behavior under normal transmission conditions and exhibits measurable deviations when adversarial interference is present. These observations validate the correctness of the simulation model and demonstrate its ability to accurately represent theoretical protocol behavior.

Overall, the experimental results confirm that both QBER and key generation rate serve as reliable indicators for evaluating communication reliability and detecting unauthorized interception in quantum key distribution systems.

8. Conclusion

This project presented a simulation-based implementation and performance evaluation of the BB84 quantum key distribution protocol under varying channel noise conditions and adversarial scenarios. The objective of the study was to investigate how noise and eavesdropping influence the reliability and security of quantum key exchange using classical simulation techniques.

The implemented simulation model successfully reproduced the fundamental workflow of the BB84 protocol, including random bit generation, basis selection, transmission through a quantum channel, measurement by the receiver, basis comparison through a classical channel, and error estimation using QBER. The inclusion of an intercept–resend attacker allowed the study to evaluate protocol behavior under realistic security threats.

Experimental results demonstrated that in the absence of an attacker, the Quantum Bit Error Rate increases proportionally with channel noise. This confirms that the protocol behaves predictably under normal communication conditions and that transmission errors primarily result from environmental disturbances. However, when an eavesdropper is introduced, the QBER increases significantly even at zero noise levels. The observed baseline error rate of approximately 25 percent matches theoretical expectations and validates the effectiveness of the simulation model.

The study also evaluated the key generation rate as an additional performance metric to measure communication efficiency. The results showed that the key generation rate decreases as channel noise increases and decreases further in the presence of adversarial interference. This indicates that both noise and interception negatively affect the number of usable bits shared between communicating parties.

These findings highlight one of the most important characteristics of the BB84 protocol: its ability to detect unauthorized interception through measurable disturbances in the transmitted quantum states. By monitoring QBER values, Alice and Bob can determine whether the communication channel is secure or compromised.

Overall, the results confirm that simulation-based modeling provides an effective approach for understanding the behavior of quantum cryptographic protocols and analyzing their performance under different communication scenarios. The successful implementation of the BB84 protocol in

this study demonstrates its robustness as a secure key distribution mechanism and validates its importance in future secure communication systems.

9. Future Scope

Although the current simulation provides a detailed analysis of the BB84 protocol under noise and intercept–resend attack conditions, several opportunities exist for extending this work to improve realism and broaden the scope of analysis.

One possible extension involves the implementation of more advanced attack strategies beyond the intercept–resend model. For example, photon-number-splitting attacks and beam-splitting attacks can be simulated to evaluate protocol performance against more sophisticated adversarial techniques. These attack models more closely represent real-world threats in practical quantum communication systems and would provide deeper insight into protocol robustness.

Another important direction for future work is the comparative analysis of different quantum key distribution protocols. Protocols such as B92 and E91 offer alternative approaches to secure quantum communication and can be evaluated alongside BB84 to compare their performance under identical channel conditions. Such comparisons would provide a broader understanding of the strengths and limitations of various QKD techniques.

The simulation framework developed in this study can also be extended by incorporating realistic quantum channel models that include photon loss, detector inefficiencies, and polarization drift effects. Including these factors would improve the accuracy of performance evaluation and make the simulation more representative of practical communication environments.

Further improvements may involve implementing the protocol using quantum computing simulation platforms such as Qiskit or other quantum circuit modeling tools. These platforms allow simulation of quantum gates and measurement operations more accurately than classical approximations and can provide deeper insight into quantum state behavior during transmission.

Another promising direction is the integration of error correction and privacy amplification stages into the simulation model. These processes are essential components of practical quantum key distribution systems and play an important role in ensuring the final security of the generated keys. Including these stages would enhance the completeness of the simulation and make it more suitable for real-world applications.

Finally, future research may explore the possibility of implementing the BB84 protocol using actual quantum communication hardware or optical fiber–based quantum transmission systems. Experimental validation using real devices would provide valuable insights into implementation challenges and help bridge the gap between theoretical models and practical deployment of quantum cryptographic systems.

These extensions would significantly enhance the scope of the present work and contribute toward the development of more secure and efficient quantum communication technologies in the future.

10. References

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Nielsen, M. A., & Chuang, I. L. (2010). Quantum Computation and Quantum Information. Cambridge University Press.

Scarani, V., Bechmann-Pasquinucci, H., Cerf, N., Dušek, M., Lütkenhaus, N., & Peev, M. (2009). The Security of Practical Quantum Key Distribution. Reviews of Modern Physics.

11. Appendices

11.1. Appendix A: Simulation Parameters

Number of transmitted bits: 1000

Number of trials per experiment: 30

Noise range tested: 0 to 0.20

Attack model: intercept-resend attack

11.2. Appendix B: Performance Metrics

Quantum Bit Error Rate (QBER):

QBER = Number of incorrect bits / Total sifted bits

Key Generation Rate:

Key Rate = Number of correct shared bits / Total transmitted bits

11.3. Appendix C: Graphical Results

The following plots were generated:

- Noise vs QBER (No Eve)
- Noise vs QBER (Eve present)
- Noise vs Key Generation Rate

These graphs demonstrate protocol performance under normal and adversarial communication conditions.